

Grade 6 Accelerated
Unit 2
Lesson 1 Practice

Name Answer Key
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Rewrite each decimal as an equivalent fraction. Simplify if possible.

1. 1.36

$$1 \frac{36}{100} = 1 \frac{9}{25}$$

or $\frac{136}{100} = \frac{34}{25}$

2. 0.022

$$\frac{22}{1000} = \frac{11}{500}$$

Convert each fraction to an equivalent decimal.

3. $\frac{17}{20} = \frac{85}{100} = 0.85$

4. $2\frac{1}{5} = 2\frac{20}{100} = 2.2$

Rewrite each decimal as a percentage.

5. 6.31

$$631\%$$

6. 0.007

$$0.7\%$$

Convert each percentage to an equivalent decimal.

7. 5%

0.05

8. 138%

1.38

9. Select all numbers that are not equivalent to 0.04. Show your work.

$$\frac{2}{5}$$

$$0.4\%$$

$$0.40\%$$

$$\frac{4}{100}$$

$$\frac{40}{100}$$

$$0.4$$

$$0.004$$

$$0.0040$$

$$0.04$$

$\frac{2}{5}$, 0.4% and 0.40% are not equivalent to 0.04.

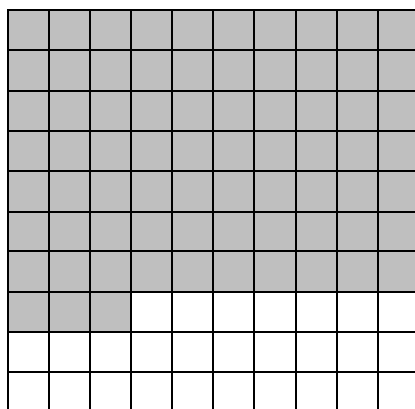
10. Emilio claims that 0.025 is equivalent to 25%. Explain whether or not he is correct.

0.025 is 2.5% when written as a percentage. Since $2.5\% \neq 25\%$, Emilio is incorrect.

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1. A model representing a numerical value is shown.



Write the numerical value represented by the model as a fraction and as a percentage.

As a Fraction	As a Percentage
$\frac{73}{100}$	73%

Convert each fraction to a percentage by generating an equivalent fraction with a denominator of 100.

	Original Fraction	Equivalent Fraction with a Denominator of 100	Percentage
Example	$\frac{17}{20}$	$\frac{17}{20} \times \frac{5}{5} = \frac{85}{100}$	85%
2.	$\frac{37}{50}$	$\frac{37}{50} \times \frac{2}{2} = \frac{74}{100}$	74%
3.	$\frac{7}{10}$	$\frac{7}{10} \times \frac{10}{10} = \frac{70}{100}$	70%
4.	$\frac{9}{4}$	$\frac{9}{4} \times \frac{25}{25} = \frac{225}{100}$	225%

Convert each fraction to a percentage and each percentage to an equivalent fraction. Simplify any fractions if possible.

	Fraction	Percentage
5.	$\frac{19}{25} \times \frac{4}{4} = \frac{76}{100}$	76%
6.	$\frac{15}{100}$ or $\frac{3}{20}$	15%
7.	$\frac{62}{100}$ or $\frac{31}{50}$	62%
8.	$\frac{6}{5} = \frac{120}{100}$	120%
9.	$\frac{125}{100}$ or $\frac{5}{4}$	125%

10. Hiro said that $\frac{3}{5}$ is equivalent to 75% because $\frac{3}{5} \times \frac{25}{25} = \frac{75}{100} = 75\%$. Explain whether you agree with Hiro.

He is wrong because $5 \times 25 = 125$. It should be 60% because $\frac{3}{5} \times \frac{20}{20} = \frac{60}{100}$.

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Lesson 3 Practice

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A fraction is shown.

$$\frac{1}{8}$$

1. Rewrite the fraction as an equivalent decimal.

$$\begin{array}{r} .1 \quad 2 \quad 5 \\ 8 \overline{) 1 . 0 0 0} \\ \underline{8} \\ 2 0 \\ \underline{-1} 6 \\ 4 0 \\ \underline{-4} 0 \\ 0 \end{array}$$

0.125

2. Based on your answer from question 1, what is an equivalent percentage to $\frac{1}{8}$?

$$0.125 = 12.5\%$$

Complete the table by rewriting each fraction as a decimal and as a percentage.

	Fraction	Decimal	Percentage
3.	$\frac{11}{40}$	$\begin{array}{r} .2 \quad 7 \quad 5 \\ 4 0 \overline{) 1 1 . 0 0} \\ \underline{-8} 0 \\ 3 0 0 \end{array}$	27.5%

		$ \begin{array}{r} -2 \quad 8 \quad 0 \\ \hline 2 \quad 0 \quad 0 \\ -2 \quad 0 \quad 0 \\ \hline 0 \end{array} $	
		0.275	
4.	$\frac{13}{8}$	$ \begin{array}{r} 1 \quad .6 \quad 2 \quad 5 \\ 8 \overline{) 1 \quad 3 \quad .0 \quad 0 \quad 0} \\ \underline{-8} \\ 5 \quad 0 \\ \underline{-4 \quad 8} \\ 2 \quad 0 \\ \underline{-1 \quad 6} \\ 4 \quad 0 \\ \underline{-4 \quad 0} \\ 0 \end{array} $	162.5%
		1.625	

Complete the table by rewriting each value in its equivalent forms. Simplify any fractions if possible.

	Fraction	Decimal	Percentage
5.	$\frac{88}{1000} = \frac{11}{125}$	0.088	8.8%
6.	$1\frac{75}{100} = 1\frac{3}{4}$ or $\frac{7}{4}$	1.75	175%
7.	$2\frac{9}{20}$	2.45	245%
8.	$3\frac{15}{100} = 3\frac{3}{20}$	3.15	315%
9.	$\frac{64}{1000}$	0.064	6.4%

10. Determine which of the numerical values is not equivalent to the others. Explain your reasoning.

7.2

7.2 is greater than 0.72

$\frac{18}{25}$
↓
0.72

72%
↓
0.72

0.72

$\frac{72}{100}$
↓
0.72

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Lesson 4 Practice

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Identify which mathematical operation would be used to answer each question. Provide an explanation or sketch to support your response.

Information	Operation	Explanation/Sketch
1. In the 2020-2021 season, Florida produced 52.5 million boxes of oranges and 4.2 million boxes of grapefruits. How many times more boxes of oranges were produced than grapefruits?	$\div$	$52.5 \div 4.2$ will tell how many times more
2. Marisol bought $4\frac{3}{8}$ yards of ribbon to make necklaces and another $2\frac{1}{5}$ yards of ribbon for bracelets. How much more ribbon did Marisol buy for necklaces than for bracelets?	$-$	Subtract $4\frac{3}{8} - 2\frac{1}{5}$ to find the difference
3. The population of the United States in 2019 was 329.10 million. If the population increased by 1.94 million in 2020, what was the population in 2020?	$+$	"Increased by" tells us to add
4. Franco and Elliot are on their school's swim team. Last week, Franco swam $5\frac{7}{10}$ miles. Elliot swam $1\frac{1}{3}$ times as far as Franco. How far did Elliot swim?	$\times$	Multiply $5\frac{7}{10}$ by $1\frac{1}{3}$

For each real-world context, write a numeric expression that could be used to calculate the answer, then show your work and write your answer using the correct units.

Information	Numeric Expression	Work and Answer with Correct Units
5. In the 2020-2021 season, Florida produced 52.5 million boxes of oranges and 4.2 million boxes of grapefruits. How many times more boxes of oranges were produced than grapefruits?	$\begin{array}{r} 52.5 \\ 4.2 \end{array}$	$\begin{array}{r} 4 \quad .2 \overline{) 52.5} \\ \underline{4 \quad 2} \\ 1 \quad 0 \quad 5 \\ \underline{8 \quad 4} \\ 2 \quad 1 \quad 0 \\ \underline{-2 \quad 1 \quad 0} \\ 0 \end{array}$

		12.5 times more oranges
6. Marisol bought $4\frac{3}{8}$ yards of ribbon to make necklaces and another $2\frac{1}{5}$ yards of ribbon for bracelets. How much more ribbon did Marisol buy for necklaces than for bracelets?	$4\frac{3}{8} - 2\frac{1}{5}$	$\frac{35}{8} - \frac{11}{5} = \frac{175}{40} - \frac{88}{40} = \frac{87}{40} = 2\frac{7}{40}$
7. The population of the United States in 2019 was 329.10 million. If the population increased by 1.94 million in 2020, what was the population in 2020?	$329.10 + 1.94$	$329.10 + 1.94 = 331.04$
8. Franco and Elliot are on their school's swim team. Last week, Franco swam $5\frac{7}{10}$ miles. Elliot swam $1\frac{1}{3}$ times as far as Franco. How far did Elliot swim?	$5\frac{7}{10} \cdot 1\frac{1}{3}$ or $\frac{57}{10} \cdot \frac{4}{3}$	$\frac{228}{30} = \frac{38}{5}$ or $7\frac{3}{5}$

For gardening, Hector has a large water jug that holds $5\frac{3}{4}$ quarts and a small water jug that holds $2\frac{1}{3}$ quarts.

9. Which of the questions could be answered using division?
- How much more can the large water jug hold than the small water jug?
 - If Hector fills both jugs with water, how much water will he have?
 - If Hector fills the small jug with water 4 times, how much total water will that be?
 - ✓ Suppose Hector uses the small jug to fill the large jug. How many times will he have to fill the small jug?
10. Using the question that you chose in number 9, write a numeric expression and answer the question. Show your work and include correct units.

$$5\frac{3}{4} \div 2\frac{1}{3} = \frac{23}{4} \div \frac{7}{3} = \frac{23}{4} \cdot \frac{3}{7} = \frac{69}{28} \text{ or } 2\frac{13}{28} \text{ times}$$

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Lesson 5 Practice

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1. A list of financial terms is shown.

taxes income deposit costs withdrawal return on investment expenses

Explain which financial terms are associated with subtraction and why.

Taxes, costs, withdrawal, expenses all come out of a person's money.

Samir works at a café. He makes \$18.55 per hour. Last week, Samir worked 22 hours and paid \$43.62 in taxes.

2. What was Samir's gross income last week? Show your work.

$$18.55 \times 22 = \$408.10$$

3. What was Samir's net income last week? Show your work.

$$408.10 - 43.62 = \$364.48$$

4. Suppose Samir also earned \$78.32 in tips last week, what is his net income then?

$$364.48 + 78.32 = \$442.80$$

5. Marshawn has \$3,857.28 in his checking account. If he withdraws \$728.50, what will his new account balance be?

$$3857.28 - 728.50 = \$3128.78$$

6. Aria has \$985.98 in her savings account. She has a check to deposit for \$247.84. What will her new account balance be after she deposits the check?

$$985.98 + 247.84 = \$1233.82$$

Hamilton is on the committee to plan his middle school dance. Their budget is shown.

Item	Amount
Ticket price in advance	\$3.75
Ticket price at the door	\$5.00
Fee paid to DJ	\$223.50
Cost to purchase decorations	\$37.89
Cost to purchase snacks	\$40.36
Snack price at the dance per item	\$1.00

7. Complete the table by listing each item from the budget as part of the gross income or part of the expenses.

Part of Gross Income	Part of Expenses
Ticket price in advance Ticket price at door Snack prices at the dance	Fee to DJ Cost for decorations Cost for snacks

8. The Dance Committee sold 172 tickets in advance. What was their income from advance ticket sales?

$$172 \times 3.75 = \$645$$

9. What was the amount of the Dance Committee's total expenses?

$$223.50 + 37.89 + 40.36 = \$301.75$$

10. Suppose their gross income was \$632.50. What was the Committee's net income?

$$632.50 - 301.75 = \$330.75$$

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Ariana works as a barista at The Ground Bean. On Saturday, she worked 6.5 hours, and she earned \$16.80 per hour. Over the course of the day on Saturday, she also made \$32.75 in tips.

1. What operations are needed to determine the total amount of money Ariana made at her job on Saturday?

Multiplication and addition

2. Write a numeric expression to determine the total amount of money Ariana made at her job on Saturday.

$$6.5(16.80) + 32.75$$

3. Determine the total amount of money Ariana made at her job on Saturday.

$$109.20 + 32.75 = \$141.95$$

Ms. Hernandez is making a big batch of hot chocolate. She mixes $3\frac{2}{3}$ cups of melted chocolate with $5\frac{1}{4}$ cups of hot milk. Each serving of hot chocolate is $\frac{5}{6}$ cups.

4. What operations are needed to determine the number of servings Ms. Hernandez is making?

Addition and division

5. Write a numeric expression that can be used to determine the number of servings Ms. Hernandez is making.

$$(3\frac{2}{3} + 5\frac{1}{4}) \div \frac{5}{6}$$

6. Gillian went to the grocery store and bought 2.4 pounds of hamburger meat for \$6.95 per pound. She paid with a \$20 bill. How much change did she receive?

$$20 - 2.4 \times 6.95$$

$$20 - 16.68 = \$3.32$$

7. Suppose Gillian thought she should get \$10.65 in change because $2.4 + 6.95 = 9.35$ and $20 + 9.35 = 10.65$. Explain to Gillian why she is incorrect.

Answers may vary. 2.4 is not money, it is the weight of the meat.

8. For the first 15 days in January, there was a total of $8\frac{3}{8}$ inches of snow in Chicago. On the 16th, it snowed $\frac{2}{5}$ of an inch every hour for 4 hours. What was the total snowfall for the first 16 days in January?

$$8\frac{3}{8} + 4 \times \frac{2}{5}$$

$$\frac{67}{8} + \frac{8}{5} \rightarrow \frac{335}{40} + \frac{64}{40} = \frac{399}{40} = 9\frac{39}{40} \text{ inches}$$

9. Explain why your answer to question 8 makes sense.

Answers will vary.

10. Marcello bought a plant that was 5.24 centimeters tall. If it grows approximately 0.22 centimeters per day, how many days will it be before the plant is 7 centimeters tall?

$$7 - 5.24 = 1.76$$

$$1.76 \div .22 = 8 \text{ days}$$

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Lesson 7 Practice

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1. Sara bought 8.6 yards of fabric that cost \$12.40 per yard. If she had a coupon for \$10 off, how much did Sara spend on the fabric altogether?

$$8.6 \times 12.40 = 106.64$$

$$106.64 - 10 = \$96.64$$

Francis is buying wallpaper for 2 walls in his bedroom. The first wall measures 9.2 feet by 7.5 feet, and the second wall measures 9.2 feet by 8.8 feet. Both walls are rectangular. Wallpaper costs \$7.40 per square foot, and a bucket of primer costs \$28.99.

2. How many square feet of wallpaper does Francis need altogether?

$$9.2 \times 7.5 + 9.2 \times 8.8 = 69 + 80.96 = 149.96 \text{ sq ft}$$

3. How much will Francis spend on wallpaper and one bucket of primer?

$$7.40(149.96)$$

$$1109.70 + 28.99 = \$1138.69$$

Angelica's goal is to run $3\frac{1}{4}$ miles on Monday, $2\frac{5}{12}$ miles on Tuesday, and $3\frac{5}{6}$ miles on Wednesday. She runs on a track where each lap is $\frac{1}{3}$ of a mile.

4. How many total laps will Angelica have to run to meet her goal for Monday thru Wednesday?

$$\left(3\frac{1}{4} + 2\frac{5}{12} + 3\frac{5}{6}\right) \div \frac{1}{3}$$

$$\left(9\frac{1}{2}\right) \div \frac{1}{3}$$

$$\frac{19}{2} \cdot \frac{3}{1} = \boxed{\frac{57}{2} \text{ or } 28\frac{1}{2} \text{ laps}}$$

5. Suppose last week Angelica ran a total of 40 laps. How many more miles will she have to run after Wednesday to run the same distance this week as last week?

$$40 - 28\frac{1}{2} = 11\frac{1}{2} \text{ laps} \left(\frac{1}{3}\right)$$

$$\frac{23}{2} \cdot \frac{1}{3} = \frac{23}{6} = \boxed{3\frac{5}{6} \text{ miles}}$$

Gustav is saving his money to buy a new computer. He currently has \$318.73 in his savings account. The table shows the occasional jobs Gustav works to earn money.

Job	Rate
Lawn-mowing	\$12.50 per lawn
Car-washing	\$13.25 per car
Babysitting	\$16 per hour
DJing	\$21 per hour

6. On Saturday, Gustav mowed 3 lawns and washed 7 cars. How much money did he make altogether on Saturday?

$$(3 \times 12.50 + 7 \times 13.25) \rightarrow 37.50 + 92.75 = \$130.25$$

7. On Sunday, Gustav babysat his cousin for 3.5 hours. Then, he DJed a party that was 2.5 hours long. How much money did Gustav make on Sunday?

$$(3.5 \times 16 + 2.5 \times 21) \rightarrow 56 + 52.50 = \$108.50$$

8. If he deposits all his income from the weekend into his savings account, how much will Gustav now have in his savings account?

$$318.73 + 130.25 + 108.50 = \$557.48$$

9. Suppose the computer Gustav wants to buy costs \$999. How much more does he need to save?

$$999.00 - 557.48 = \$441.52$$

10. What is a combination of jobs that Gustav could work such that he would have enough money saved to buy his computer?

Answers will vary.